United States Patent	12403431
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Tamminga; Jakob et al.

Coned auger

Abstract

Coned augers for bulk mixers are described. The coned augers comprise first end and second end an auger post having a first end opposite a second end; fighting extending along and about the auger post, the pitch oriented to move bulk material generally from the first end toward the second end during operation of the auger; a coned section along the auger post, the coned section comprising a first cone protruding about the periphery of the auger post, the first cone having a tapered end opposite a wide end. Mixers comprising the coned augers are also described.

Inventors: **Tamminga; Jakob (East Garafraxa, CA), Bouwman; Nathan (Fergus, CA)**

Applicant: **JAY-LOR INTERNATIONAL INC.** (East Garafraxa, CA)

Family ID: **96876757**

Appl. No.: **17/673060**

Filed: **February 16, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20230256402 A1	Aug. 17, 2023

Related U.S. Application Data

us-provisional-application US 63151180 20210219

Publication Classification

Int. Cl.: **B01F27/192** (20220101); **B01F27/1145** (20220101); **B01F27/117** (20220101); **B01F27/722** (20220101); B01F101/18 (20220101); B01F101/33 (20220101)

U.S. Cl.:

CPC **B01F27/192** (20220101); **B01F27/1145** (20220101); **B01F27/117** (20220101); **B01F27/722** (20220101); B01F2101/18 (20220101); B01F2101/33 (20220101); B01F2215/0422 (20130101)

Field of Classification Search

CPC: B01F (2101/33); B01F (2101/18); B01F (2215/0422); B01F (27/722); B01F (27/1145); B01F (27/117)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2813704	12/1956	Mackissic	198/312	B01F 33/5023
3175737	12/1964	Schnell	222/274	A01C 15/16
3244271	12/1965	Wenning	198/669	A01D 61/002
3263276	12/1965	Maier	425/208	B29C 45/50
3421740	12/1968	Behrens	366/186	B01F 27/726
3702128	12/1971	Trotter	198/582	B65G 53/08
3728053	12/1972	Stillhard	264/DIG.83	B29C 45/50
4453375	12/1983	Field	56/364	A01D 61/008
5356054	12/1993	Loppoli	241/101.76	B01F 27/726
5385403	12/1994	Knight	366/603	A01C 3/063
5622323	12/1996	Krueger	241/101.76	B01F 33/5023
5967657	12/1998	West	366/195	A01K 5/002
5984218	12/1998	Peat	241/294	B01F 27/726
6328919	12/2000	Pham	425/208	B29C 48/76
11426701	12/2021	Gruber	N/A	B29B 7/484
2014/0001678	12/2013	Hara	264/328.14	B29C 45/60
2019/0270057	12/2018	Tamminga	N/A	B01F 33/502

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
877683	12/1970	CA	N/A
1596652	12/2005	EP	A01F 29/005
WO-2009154698	12/2008	WO	B29C 33/76

OTHER PUBLICATIONS

Zareiforoush, H., Komarizadeh, M. H., & Alizadeh, M. R. (2010). A. Review on screw conveyors performance evaluation during handling process. J. Sci. Rev, 2, 55-63. (Year: 2010). cited by examiner

---

Primary Examiner: Wang; Claire X

Assistant Examiner: McCarty; Patrick M

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to U.S. provisional application 63/151,180, filed Feb. 19, 2021, the entire contents of which are hereby incorporated by reference.

FIELD OF THE INVENTION
(1) The invention relates to agricultural machinery and more specifically to horizontal augers and bulk mixers comprising a coned auger for use in mixing bulk material such as feed, compost, animal bedding, fertilizer, etc.
BACKGROUND OF THE INVENTION
(2) Horizontal mixers are typically used to mix bulk material such as agricultural material including feed and bedding. Conventional mixers typically include one or more augers positioned horizontally and operate at about 5-12 rpm to mix the bulk material. Generally, a significant amount of horsepower is required to rotate the augers as the load to be mixed can be very large and can range for example from 2,000 to 40,000 pounds.
(3) Further, the amount of time required to sufficiently mix the material can be significant and can be frustrating for the user as it can take up to 8-10 minutes to sufficiently mix a load or longer.
(4) The horizontally positioned augers may be ribbon augers or solid fighting augers having a fighting positioned away from a central auger post. The fighting typically has a pitch oriented to draw material away from the ends of the mixer toward the center and typically to a conversion zone or spot which generally aligns with a door in the side of the mixer. Operation of the traditional auger in a mixer such as a horizontal mixer can result in compressing, clumping of the mix and/or the break down or destruction or reduction of effective fiber in the feed. Dead spaces may also be generated in the mixer during mixing. Uneven unloading of the mixer may also be observed.
(5) A need therefore exists for an auger for use in a mixer, for example a horizontal mixer, that mitigates any of the downsides referred to above or as observed in the industry.
SUMMARY OF THE INVENTION
(6) In one embodiment, the present invention provides for a coned auger for a bulk mixer, the coned auger comprising: an auger post having a first end opposite a second end; fighting extending along and about the auger post, the pitch oriented to move bulk material generally from the first end toward the second end during operation of the auger; a coned section along the auger post, the coned section comprising a first cone protruding about the periphery of the auger post, the first cone having a tapered end opposite a wide end.
(7) In a further embodiment of the coned auger or coned augers as outlined above, the auger is a ribbon auger and the fighting is ribbon fighting.
(8) In a further embodiment of the coned auger or coned augers as outlined above, the first cone is generally positioned substantially at or near the second end of the auger post, the tapered end oriented toward the first end of the auger post and the wide end oriented toward the second end of the auger post.

(9) In a further embodiment of the coned auger or coned augers as outlined above, the coned section further comprises a second cone protruding from the auger post comprising a tapered end opposite a wide end and wherein the second cone has an orientation opposite to the first cone and wherein a wide end of the second cone abuts or is adjacent to the wide end of the first cone.
(10) In a further embodiment of the coned auger or coned augers as outlined above, the fighting or ribbon fighting comprises a first pitch section comprising a pitch oriented to move bulk material generally from the first end toward the second end and a second pitch section situated opposite the first pitch section and having an orientation to move bulk material generally from the second end toward the first end of the auger, and wherein the position at which the first pitch section meets the second pitch section defines a conversion point, and wherein the coned section overlaps with the conversion point.
(11) In a further embodiment of the coned auger or coned augers as outlined above, the auger or augers further comprise plating spanning the distance from the ribbon fighting to the first, and optionally second cone, in the coned section thereby forming solid auger fighting in the coned section.
(12) In a further embodiment of the coned auger or coned augers as outlined above, the plating is integrated into the fighting in the coned section and the coned section comprises solid fighting.
(13) In a further embodiment of the coned auger or coned augers as outlined above, the auger or augers further comprise auger paddles mounted to the auger for aiding unloading of the mixing chamber.
(14) In a further embodiment of the coned auger or coned augers as outlined above, the first and/or second cone is a round cone or pyramidal cone of 3 or more sides.
(15) In a further embodiment of the coned auger or coned augers as outlined above, a diameter of the wide end of the first or second cone is less than a diameter of fighting.
(16) In a further embodiment of the coned auger or coned augers as outlined above, the first cone and/or the second cone have a generally frustum shape.
(17) In a further embodiment of the coned auger or coned augers as outlined above, the first cone and the second cone have a different taper angle.
(18) In a further embodiment of the coned auger or coned augers as outlined above, a taper angle of the first cone is 135° or greater, optionally 140° or greater, optionally 145° or greater, optionally 150° or greater, optionally 155° or greater, optionally 160° or greater, optionally 165° or greater, optionally 170° or greater or optionally 175° or greater.
(19) In a further embodiment of the coned auger or coned augers as outlined above, the pitch of the fighting or ribbon fighting is at least the same as the diameter of the fighting or ribbon fighting.
(20) In a further embodiment of the coned auger or coned augers as outlined above, the pitch of the fighting or ribbon fighting is about 150 percent the diameter of the fighting or ribbon flighting.
(21) In another embodiment, the present invention provides for a bulk mixer for mixing a bulk material, the bulk mixer comprising:
(22) a mixing chamber for containing the bulk material to be mixed;
(23) an exit door for expelling mixed bulk material from the mixing chamber; and
(24) a coned auger in the mixing chamber, the coned auger as defined herein.
(25) In a further embodiment of the mixer or mixers as outlined above, the coned auger is as defined herein, wherein the conversion point is generally in line with the outlet or overlaps with the outlet in the mixing chamber and the coned section is generally in line with the exit door.
(26) In a further embodiment of the mixer or mixers as outlined above, the bulk mixer is a horizontal mixer.
(27) In yet another embodiment, the present invention provides for a horizontal mixer for mixing bulk material, the horizontal mixer comprising:
(28) a mixing chamber for accommodating bulk material to be mixed, the mixing chamber comprising: side walls connected at their bottom edges by a trough; a front wall spanning a front

side of the container; a rear wall spanning a rear side of the container; and an exit door through which mixed material can exit; one or more bottom augers situated longitudinally in the trough, wherein at least one of the one or more bottom augers is a coned auger as defined herein; and one or more top augers situated longitudinally in the mixing container substantially above the one or more bottom augers; wherein the trough has a shape to accommodate the one or more bottom augers; and optionally a taper angle of the first cone is 135° or greater, optionally 140° or greater, optionally 145° or greater, optionally 150° or greater, optionally 155° or greater, optionally 160° or greater, optionally 165° or greater, optionally 170° or greater or optionally 175° or greater.

(29) In a further embodiment of the mixer or mixers as outlined above, the fighting or ribbon fighting comprises a first pitch section comprising a pitch oriented to move bulk material generally from the front wall toward the exit door and a second pitch section situated opposite the first pitch section and having an orientation to move bulk material generally from the back wall toward the exit door of the auger, and wherein the position wherein the first pitch section meets the second pitch section defines a conversion point, and wherein the coned section overlaps with the conversion point.

(30) In a further embodiment of the mixer or mixers as outlined above, the conversion point is generally in line with the exit door or overlaps with the exit door in the mixing chamber and the coned section is generally in line or overlaps with the exit door.

(31) In a further embodiment of the mixer or mixers as outlined above, there are at least two bottom augers and the at least two bottom augers are coned augers, each independently as defined in herein, the at least two bottom augers being defined as an exit door side bottom auger situated on the side of the mixing chamber comprising the exit door and a non-exit door side bottom auger situated on the side of the mixing chamber that does not comprise the exit door.

(32) In a further embodiment of the mixer or mixers as outlined above, the exit door side bottom auger has the coned section positioned to overlap with the exit door.

(33) In a further embodiment of the mixer or mixers as outlined above, the wide end of the first or second cone of the exit door side bottom auger has a smaller diameter than the wide end of the first or second cone of the non-exit door side bottom auger.

(34) In a further embodiment of the mixer or mixers as outlined above, the wide end of the first or second cone of the exit door side bottom auger has a larger diameter than the wide end of the first or second cone of the non-exit door side bottom auger.

(35) In a further embodiment of the mixer or mixers as outlined above, the coned section of the non-exit door side bottom auger is not in-line with the coned section of the exit door side bottom auger.

(36) In a further embodiment of the mixer or mixers as outlined above, the coned section of the non-exit door side bottom auger is in-line with the coned section of the exit door side bottom auger.

(37) In a further embodiment of the mixer or mixers as outlined above, the coned section of the non-exit door side bottom auger overlaps with the coned section of the exit door side bottom auger.

(38) In a further embodiment of the mixer or mixers as outlined above, the coned section of the exit door side bottom auger is comprised of a first and/or second cone having a different taper angle than a first and/or second cone of the non-exit door side bottom auger.

(39) In a further embodiment of the mixer or mixers as outlined above, the one or more top augers is a ribbon auger.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a top schematic view illustrative of one embodiment of a coned ribbon auger according to one embodiment of the invention.

(2) FIG. **2** is an isometric view of the coned ribbon auger shown in FIG. **1**.
(3) FIG. **3** is a top schematic view illustrative of another embodiment of a coned ribbon auger according to one embodiment of the invention.
(4) FIG. **4** is an isometric view of the coned ribbon auger shown in FIG. **3**.
(5) FIG. **5** is a top schematic view illustrative of another embodiment of a coned ribbon auger according to one embodiment of the invention.
(6) FIG. **6** is an isometric view of the coned ribbon auger shown in FIG. **5**.
(7) FIG. **7** is a top schematic view illustrative of another embodiment of a coned ribbon auger according to one embodiment of the invention.
(8) FIG. **8** is an isometric view of the coned ribbon auger shown in FIG. **7**.
(9) FIG. **9**A is an isometric view illustrative of one embodiment of a bulk mixer comprising one embodiment of a coned ribbon auger according to the invention.
(10) FIG. **9**B is a top schematic view illustrative of the bulk mixer and coned ribbon auger as shown in FIG. **9**A.
(11) FIG. **9**C is an end view illustrative of the bulk mixer and coned ribbon auger as shown in FIG. **9**A.
(12) FIG. **10** is a top view illustrative of another embodiment of a bulk mixer comprising another embodiment of a coned ribbon auger according to the invention.
(13) FIG. **11** is a top view illustrative of another embodiment of a bulk mixer comprising another embodiment of a coned ribbon auger according to the invention.

DETAILED DESCRIPTION

(14) Described herein are examples and embodiments of coned augers and horizontal mixers comprising coned augers including apparatuses, systems therefor and methods and processes of using same, for mixing bulk material such as agricultural material for example feed, compost or bedding. It will be appreciated that embodiments and examples are provided herein for illustrative purposes intended for those skilled in the art, and are not meant to be limiting in any way. All references to embodiments or examples throughout this disclosure should be considered as references to illustrative and non-limiting embodiments and illustrative and non-limiting examples. It will be appreciated that none of the features disclosed herein are intended to be essential unless specifically stipulated as such. Reference to any dimensions or measurements is for illustrative purposes and not intended to be limiting and is not intended to be an exact and limiting measurement and the term “about” is intended to be applied to all such dimensions and measurements and at least accounts for inaccuracies and error associated with taking such measurements and the devices for taking such measurements.
(15) An auger having a coned section which aids in more effectively mixing a bulk material in a bulk mixer is disclosed herein. It has been determined that adding a coned section to an auger, such as a ribbon auger or a solid auger can allow for the auger to more effectively mix a bulk material when used in a mixer, for example a horizontal mixer. Using a coned auger can potentially reduce mix times and/or the energy required to mix a bulk material, reduce clumping and/or compression of the mixing material, reduce dead spots in the mix, and/or may reduce damage to effective fibre in a feed mix.
(16) A coned auger according to the present invention may be used in a bulk mixer such as a bulk agricultural mixer, such as a bulk feed mixer and may be used in a horizontal bulk mixer such as a horizontal agricultural mixer, such as a horizontal feed mixer. An example of a horizontal mixer suitable for use with a coned auger is shown in Canadian patent application 3,034,724 filed Feb. 22, 2019 and entitled “Horizontal Mixer With Stacked Augers” incorporated herein by reference in its entirety.
(17) One example of a coned ribbon auger according to one aspect of the present invention is a coned ribbon auger shown generally at **100** in FIGS. **1** and **2**. The coned ribbon auger **100** includes an auger post **125** along which ribbon fighting **105** runs. The ribbon fighting **105** has a pitch

oriented to draw material from a first end of the auger **100** toward a second opposite end of the auger **100**. The ribbon auger **100** may alternatively have a pitch oriented to draw material from a second end of the auger **100** toward a first end of the auger **100**. The auger **100** shown in FIGS. **1** and **2** includes fighting having a first pitch section **145** oriented to draw material away from the first end of the auger **100** toward the second end of the auger **100** and a second pitch section **140** oriented to draw material away from the second end of the auger **100** toward the first end of the auger **100**. This may also be referred to as forward fighting and reverse fighting. The point where the first pitch section **145** and the second pitch section **140** meet is referred to as the conversion point **135**. When the auger **100** is situated in a mixer, the pitch is oriented to draw material away from an end wall of the mixing chamber. When the auger **100** has two pitch sections **140** and **145**, operation of the auger draws material away from each end of the mixing chamber of the mixer. The conversion point **135** may be situated such that it is in-line with or overlaps with an exit door in the mixing chamber. This is especially useful if a single auger is used, or if a single bottom auger is used in a stacked auger setup, or multiple augers are used and the exit door side auger has a conversion point that overlaps with or is in-line with the exit door. This is explained in more detail with reference to FIGS. **9**A to **11**.

(18) The coned ribbon auger **100** also includes a coned section **120** that is made up of a first cone **110** and optionally a second cone **115**. Each cone has a tapered end opposite a wide end. As shown in FIGS. **1** and **2**, the first cone **110** has an opposite orientation to the second cone **115** and their wide ends abut or are adjacent. Typically, the coned section **120** overlaps with the conversion point **135**. In one embodiment, the conversion point **135** is at the point where the first cone **110** and the second cone **115** abut one another.

(19) The coned section **120** may have cones whose diameter at their wide end is no more than the external diameter of the fighting **105** also referred to as the diameter of the auger.

(20) In the coned section **120**, the ribbon fighting **105** may include plating **130** spanning the gap between the cone **110** or **115** and the ribbon fighting, resulting is solid fighting in the coned section **120**. The plating **130** prevents material from wrapping around the cone **110** and/or **115** and getting stuck between the cones and the fighting **105** during operation of the auger **100**. During manufacture of the auger **100**, the coned section **120** may be added to the auger post **125** and then the fighting may be added. Solid fighting may be added around the coned section **120** and ribbon fighting may be added along all or substantially all of the remaining auger post. The plating **130** may be integrated directly into the fighting of the coned section **120**.

(21) It has been observed that using a coned auger such as that shown with reference to FIGS. **1** and **2** can result in mixing of a bulk material in a horizontal auger in a much reduced time while using the same amount of energy to power the mixer. A user can also reduce the amount of energy used and still obtain a reduced mix time or similar mix time if the energy is even further reduced. This can result in a reduced power consumption, reduced fuel consumption and reduced carbon impact.

(22) Furthermore, it has been observed that the coned auger appears to push feed out of the exit door in a horizontal mixer more gently and more quickly. The coned auger appears to mix the bulk material without compressing it as much as a traditional solid auger or a traditional ribbon auger and the mixed feed rolls out the door in much better shape with less damage to effective fibre.

(23) For example, when using a steam flaked corn, a mixer using a traditional ribbon auger has been observed to crumple it and generate compressed crumbs which reduces the effective fibre. Avoiding or reducing the break down or destruction of effective fibre is preferable.

(24) As can be seen in the coned section **120** in FIGS. **1** and **2**, the first cone **110** and the second cone **115** can have different lengths which results in a different taper angle between the cones **110** and **115**. The taper angle being the angle formed by the auger post **125** and the cone **110** or **115**. An example of the taper angle is shown in FIG. **3** where the second cone **315** has a taper angle of 163°. An optional taper angle of 135° or greater appears to be ideal but any suitable taper angle may be

adopted. It will be appreciated that the first and second cones **110** and **115** may have similar dimensions or may be different. A cone may be longer than the other and have a different taper angle. Alternatively, the first cone **110** may have a larger or smaller diameter across the wide end than the second cone **115**. Cones of different sizes and shapes may also be used and all cones are contemplated by the inventors and are within scope of the invention. Generally, the cones may have a frustum shape.

(25) The exterior surface or shell of the cones may have a smooth round shape about their circumference and be made from a single section or multiple sections fixed together, for example by welding. The piece or pieces may be stamped in order to obtain the desired shape. The exterior surface or shell may be formed by a plurality of flat surfaces that abut each other at an angle to form a pyramid shape having multiple faces. Each face may be a separate piece joined together or may be formed by stamping, pressing or by a combination of these or other suitable techniques.

(26) Any suitable interior frame may be used to support the exterior surface or shell. For example, one or more interior plates may be mounted to the auger post **105** to support the surface or shell so that the weight of the load to be mixed does not crush the cone. For example, as shown with reference to FIG. **3**, a series of plates **335** are used to support the cone surface or shell on the auger post **325**. The plates **335** may be made from any suitably strong material, for example steel or stainless steel. Further, although five plates are shown as being used to support the cones **310** and **315** in FIG. **3**, any suitable number or orientation of the support plates may be used and any suitable support structure may be implemented as needed.

(27) The pitch of the ribbon fighting **105** may vary and it will be appreciated that any suitable pitch may be used on the ribbon fighting **105** to mix bulk material as intended. The pitch may be very shallow or may be 100% or greater. The pitch is usually expressed as a length and can be expressed as a ratio as compared to the diameter of the ribbon auger. It has been observed that having a pitch of at least a 1:1 ratio of pitch:auger diameter, also expressed for the purposes of this disclosure as a pitch of least 100% tends to mix bulk material well. The pitch may be greater than 1:1 or greater than 100%. For example, the pitch in the first pitch section **145**, where the diameter of the ribbon auger **105** is 28 inches and the pitch is 36 inches, is about 129%. The pitch of the ribbon fighting **305** of the auger **300** shown with reference to FIG. **3** is 48 inches, or 48:42 or 114%. The pitch may be as great as 3:2 or 150% or may be more. It is contemplated that the pitch may be any suitable pitch that allows for usable mix to be generated by operation of the coned auger in a mixer. It is also contemplated that the pitch in the first pitch section **145** may be different than or the same as the pitch in the second pitch section **140**.

(28) It will be appreciated that the auger **100** may be fabricated from any suitable material or combination of material and it is not intended for the material to be limiting in any way. The auger **100** and the components therein may be made from steel or stainless steel and may be welded, stamped, cut or may be made using a press break or die or combination thereof.

(29) FIGS. **3** and **4** show another embodiment of a coned auger **300** for use in a bulk mixer. The coned auger **300** is generally similar to that shown with reference to FIGS. **1** and **2** in that the coned auger **300** comprises an auger post **325** along which ribbon fighting **305** runs. The ribbon fighting **305** has a pitch oriented to draw material away from each end and toward a conversion point **340** where opposite pitches meet. The coned auger **300** comprises a coned section **320** including a first cone **310** adjacent and oppositely oriented to a second cone **315**. In the embodiment shown with reference to FIGS. **3** and **4**, the first cone **310** and the second cone **315** have generally the same dimensions as opposed to the first and second cones of the auger **100** which had different lengths and different taper angles. In addition, the coned section **320** of the coned auger **300** is positioned somewhat inboard of the end of the auger post **325** where the coned section **120** of the coned auger **100** was generally at the end of the auger post **125**. This illustrates that the coned section **320** may be situated as any suitable point along the auger post **325** and overlapping with the conversion point **340**. The coned section **320** also includes the plating **330**

spanning the gap between the cone **310** or **315** and the ribbon fighting **305**, resulting is solid fighting in the coned section **320**.

(30) FIGS. **5** and **6** show another embodiment of a coned auger **500** which is comprised of a coned section **520** which includes only a first cone **510**. The coned auger **500** comprises an auger post **525** along which ribbon fighting **505** runs. The ribbon fighting **505** has a pitch oriented to draw material away from one end and toward the other end of the auger **500**. The auger **500** has only a first pitch and does not include a pitch designed to draw material away from the other end of the auger **500**. As a result, the coned section **510** is positioned at the end of the auger **500** and includes only a single cone **510**. There is no conversion point on the fighting **505** as there is no oppositely oriented pitch on the fighting. This illustrates that the coned section **520** may include only a single cone **510**. It will be appreciated that the coned section may be positioned at the end of the auger **500** but it is also contemplated that the single cone **510** may be placed away from the end of the auger **500** toward the center and the auger fighting **505** may include a first and a second pitch oriented to draw material away from each side and that the single cone **510** may be positioned at the conversion point of the two pitches. The coned section **520** also includes the plating **530** spanning the gap between the cone **510** and the ribbon fighting **505**, resulting is solid fighting in the coned section **520**. The cone **510** includes a support structure comprised of a series of plates **535** used to support the cone surface or shell on the auger post **525**. The plates **535** may be made from any suitably strong material, for example steel or stainless steel. Further, although three plates are shown as being used to support the cone **510** in FIG. **5**, any suitable number or orientation of the support plates may be used and any suitable support structure may be implemented as needed.

(31) FIGS. **7** and **8** show another embodiment of a coned auger **700** similar to that shown with reference to FIGS. **3** and **4**. The coned auger **700** comprises an auger post **725** along which ribbon fighting **705** runs. The ribbon fighting **705** has a pitch oriented to draw material away from each end and toward a conversion point **740** where opposite pitches meet. The coned auger **700** comprises a coned section **720** including a first cone **710** adjacent or abutting and oppositely oriented to a second cone **715**. In the embodiment shown with reference to FIGS. **7** and **8**, the first cone **710** and the second cone **715** have generally the same dimensions as opposed to the first and second cones of the auger **100** which had different lengths and different taper angles. In addition, the coned section **720** of the coned auger **700** is positioned generally central on the auger post **725** where the coned section **120** of the coned auger **100** was generally at the end of the auger post **125**. This further illustrates that the coned section **720** may be situated as any suitable point along the auger post **725** and overlapping with the conversion point **740**. The coned section **720** also includes the plating **730** spanning the gap between the cone **710** or **715** and the ribbon fighting **705**, resulting is solid flighting in the coned section **720**. The cones **710** and **715** include a support structure comprised of a series of plates **735** used to support the cone surface or shell on the auger post **725**. The plates **735** may be made from any suitably strong material, for example steel or stainless steel. Further, although five plates are shown as being used to support the cones **710** and **715** in FIG. **7**, any suitable number or orientation of the support plates may be used and any suitable support structure may be implemented as needed.

(32) It will be appreciated that although a coned ribbon auger has been exemplified and illustrated in reference to FIGS. **1** to **8**, it is within the contemplated invention that the auger may be a solid auger with a coned section and the invention is not limited to ribbon augers with a coned section.

(33) FIGS. **9**A, **9**B and **9**C show a mixer **900** for mixing a bulk material. The mixer **900** has an open top into which bulk material, such as bulk agricultural material, may be placed to be mixed. The mixing container is defined by side walls **930** and **935** that are connected at their bottom ends by a trough **960**. A front wall **950** and a rear wall **920** span each end of the mixing container thereby forming the open top mixing container, also referred to as a mixing chamber.

(34) The embodiment of the horizontal mixer **900** shown in FIGS. **9**A-**9**C is a horizontal mixer having a quad auger arrangement and includes two sets of stacked augers as can be more clearly

seen in the top view of FIG. **9**B and end view of FIG. **9**C. The mixer **900** includes bottom coned ribbon augers **910** which are longitudinally positioned in the mixing container. The trough **960** has a W-shape to accommodate the dual bottom coned ribbon augers **910** such that each of the bottom coned ribbon augers is at least somewhat encapsulated or sunk into its own trough.

(35) As can be seen in FIG. **9**C, the W-shaped trough **960** is comprised of two parallel troughs each adapted to accommodate its own auger and may be of different sizes based, for example, on the diameter of the auger which is to be accommodated. For example, each bottom coned ribbon auger may have a different diameter and may be positioned at a different height. It will be appreciated that any suitable auger setup may be used and the coned solid and/or ribbon augers of the present invention as defined herein may be adapted to be accommodated by a mixer, or the mixer may be adapted to accommodate a coned solid and/or ribbon auger such as those defined herein. The horizontal mixer description with reference to FIGS. **9**A-**9**C is only one example of a mixer that can use a coned auger or coned ribbon auger of the present invention and is intended to be for illustrative purposes and it is not the intention of the inventors that only horizontal mixers such as that shown in FIG. **9**A-**9**C or **10** and **11** may be used in conjunction with the coned ribbon augers of the invention.

(36) As can be seen in FIGS. **9**A-**9**C, the bottom augers of the horizontal mixer are coned ribbon augers **910**. The top augers **940** may be ribbon augers or solid augers. They may also be coned augers.

(37) The mixer **900** comprises an exit door **970** situated at any appropriate position to allow output of mixed material from the mixing chamber. In one embodiment, the exit door **970** may in the side wall **935** of the mixer **900** and may be on the driver side of the mixer **900** thereby allowing a user operating the vehicle towing the mixer to more easily see the exit door **970** in the rear-view mirror and more efficiently exit the cab of the vehicle towing the mixer to access the exit door **970**.

(38) As outlined above, the coned ribbon auger on the exit door side **911** has the coned section **990** in-line with or overlapping the exit door **970**. The non-exit door side coned ribbon auger **912** has the coned section **980** located at any suitable point along the auger post **921**.

(39) The pitch of the bottom augers **910** may be oriented to draw material away from the front wall **950** and the rear wall **920** toward the center of the mixing chamber. The conversion point of each of the bottom auger **910** may be the same or different as shown in FIGS. **9**A to **9**C.

(40) In the mixer **900** shown with reference to FIGS. **9**A-**9**C the bottom augers **910** may be any suitable coned solid or ribbon auger as described herein or as contemplated by the inventors. The bottom augers **910** shown in FIGS. **9**A-**9**C are a ribbon auger as described with reference to FIGS. **1-2** (bottom ribbon auger **911**) and FIGS. **3-4** (bottom ribbon auger **912**). The bottom augers **911** and **912** have different diameters, but, it is contemplated that the bottoms augers may have the same diameter and may optionally be have the same positioning of the coned section.

(41) FIGS. **10** and **11** show mixers having two other setups for the bottom coned ribbon augers.

(42) FIG. **10** shows a horizontal mixer **1000** comprising bottom coned ribbon augers including an exit door side coned ribbon auger **1011** and a non-exit door side coned ribbon auger **1012**. The coned ribbon auger **1011** has a coned section **1090** which overlaps with the exit door **1070** in the mixer **1000**. The coned ribbon auger **1012** has a coned section **1080** positioned along the auger post at a point which does not overlap with the coned section **1090** of the coned ribbon auger **1011**. The pitch of the bottom augers **1011** and **1012** may be oriented to draw material away from the front wall **1050** and the rear wall **1020** toward the center of the mixing chamber. The top augers **1040** may be ribbon augers or solid augers. They may also be coned ribbon augers.

(43) FIG. **11** shows a horizontal mixer **1100** comprising bottom coned ribbon augers including an exit door side coned ribbon auger **1111** and a non-exit door side coned ribbon auger **1112**. The coned ribbon auger **1111** has a coned section **1190** which overlaps with the exit door **1170** in the mixer **1100**. The coned ribbon auger **1112** has a coned section **1180** positioned at a front end of the auger post at a point adjacent the front end of the mixing chamber **1150**. The coned section **1180**

comprises a single cone such as that described with reference to FIGS. **5-6**. The pitch of the bottom auger **1111** may be oriented to draw material away from the rear wall **1120** and the front wall **1150** toward the center of the mixing chamber. The pitch of the coned ribbon auger **1112** is oriented to draw matter in a single direction, in this case from the back of the mixer **1120** toward the front **1150**. The top augers **1140** may be ribbon augers or solid augers. They may also be coned ribbon augers.
(44) The exit door side coned ribbon augers and non-exit door side coned ribbon augers described herein may have identical or non-identical diameters. In embodiments with non-identical diameters, the diameter of the exit door side coned ribbon auger may be smaller than the diameter of the non-exit door side coned ribbon auger. In other embodiments, the diameter of the exit door side coned ribbon auger may be larger than the diameter of the non-exit door side coned ribbon auger.
(45) It will be appreciated that any suitable horizontal mixer may be used with any suitable auger set up and orientation and any suitable number of augers. The mixer may comprise a coned ribbon auger, such as one of those defined herein or as contemplated by the inventors, and may have the coned section overlap with or inline with the exit door of the mixer.
(46) It will be appreciated that the coned auger as described herein may be used in conjunction with paddles mounted to the auger. The paddles may be mounted in any suitable manner and/or position. The paddles may function to aid in mixing and/or unloading of the bulk material.
(47) One or more illustrative embodiments have been described by way of example. It will be understood to persons skilled in the art that a number of variations and modifications can be made without departing from the scope and spirit of the invention as defined herein and in the claims. Such modifications and variations are within the intended scope of the invention and the contemplation of the inventors.

Claims

1. A coned auger for a horizontal feed mixer, the coned auger comprising: an auger post having a first end opposite a second end; flighting extending along and about the auger post, with a first pitch section comprising a pitch oriented to move bulk material generally from the first end toward the second end during operation of the auger; a coned section along the auger post, the coned section comprising a first cone protruding about the periphery of the auger post, the first cone having a tapered end opposite a wide end; wherein the flighting comprises ribbon flighting; wherein the pitch of the flighting is greater than a diameter of the flighting; and wherein the coned section further comprises a second cone protruding from the auger post comprising a tapered end opposite the wide end of the first cone and wherein the second cone has an orientation opposite to the first cone and wherein a wide end of the second cone abuts or is adjacent to the wide end of the first cone; and wherein a taper angle of the first cone and a taper angle of the second cone are different.
2. The coned auger of claim 1, wherein the first cone is positioned substantially at or near the second end of the auger post, the tapered end of the first cone oriented toward the first end of the auger post and the wide end oriented toward the second end of the auger post.
3. The coned auger of claim 1, wherein the flighting further comprises a second pitch section situated opposite the first pitch section and having an orientation to move bulk material generally from the second end of the auger post toward the first end of the auger post, and wherein a position at which the first pitch section meets the second pitch section defines a conversion point, and wherein the coned section overlaps with the conversion point.
4. The coned auger of claim 1, further comprising auger paddles mounted to the auger for aiding unloading of a mixing chamber.
5. The coned auger of claim 1, wherein the first and/or second cone is a round cone or pyramidal

cone of 3 or more sides.
6. The coned auger of claim 1, wherein a diameter of the wide end of the first or second cone is less than said diameter of the flighting.
7. The coned auger of claim 1, wherein the first cone and the second cone have a frustum shape.
8. The coned auger of claim 1, wherein a said taper angle of said first cone is 135° or greater.
9. The coned auger of claim 1, wherein said taper angle of the first cone is −175° or greater.
10. The coned auger of claim 1, further comprising plating spanning a gap between the cone section and the ribbon flighting thereby forming solid auger flighting in the coned section.
11. The coned auger of claim 10, wherein the plating is integrated into the flighting in the coned section.
12. A horizontal mixer for mixing feed material, the horizontal mixer comprising: a mixing chamber for accommodating bulk material to be mixed, the mixing chamber comprising: side walls connected at their bottom edges by a trough; a front wall spanning a front side of said mixing chamber; a rear wall spanning a rear side of said mixing chamber; and an exit door through which mixed material can exit; one or more bottom augers situated longitudinally in the trough, wherein at least one of the one or more bottom augers is a coned auger comprising: an auger post having a first end opposite a second end; flighting extending along and about the auger post, with a first pitch section comprising a pitch oriented to move bulk material generally from the first end toward the second end during operation of the auger; a coned section along the auger post, the coned section comprising a first cone protruding about a periphery of the auger post, the first cone having a tapered end opposite a wide end; wherein the flighting comprises ribbon flighting; wherein the pitch of the flighting is greater than a diameter of the flighting; wherein the coned section further comprises a second cone protruding from the auger post comprising a tapered end opposite the wide end of said first cone and wherein the second cone has an orientation opposite to the first cone and wherein a wide end of the second cone abuts or is adjacent to the wide end of the first cone; and wherein a taper angle of the first cone and a taper angle of the second cone are different; and at least one top auger situated longitudinally in the mixing chamber substantially above the at least one bottom auger, wherein the trough has a shape to accommodate the at least one bottom auger.
13. The horizontal mixer of claim 12, wherein said flighting comprises said first pitch section which is oriented to move bulk material generally from the front wall toward the exit door and a second pitch section situated opposite the first pitch section and having an orientation to move bulk material generally from the rear wall toward the exit door of the mixing chamber, and wherein a position wherein the first pitch section meets the second pitch section defines a conversion point, and wherein the coned section overlaps with the conversion point.
14. The horizontal mixer of claim 12, wherein one of said cones has said taper angle of at least 135°.
15. The horizontal mixer of claim 12, wherein the coned auger is an exit door side bottom auger situated on the side of the mixing chamber comprising the exit door and the horizontal mixer further comprises a non-exit door side bottom auger situated on the side of the mixing chamber opposite the exit door side; wherein the non-exit door side bottom auger is also a coned auger.
16. The horizontal mixer of claim 15, wherein the wide end of the first or second cone of the exit door side bottom auger and a wide end of a first or second cone of the non-exit door side bottom auger have non-identical diameters.
17. The horizontal mixer of claim 15, wherein a coned section of the non-exit door side bottom auger overlaps with the coned section of the exit door side bottom auger.
18. The horizontal mixer of claim 15, wherein the coned section of the exit door side bottom auger is comprised of the first and/or second cone having a different taper angle than a first and/or second cone of the non-exit door side bottom auger.
19. A coned auger for a horizontal feed mixer, the coned auger comprising: an auger post having a first end opposite a second end; flighting extending along and about the auger post, with a first

pitch section comprising a pitch oriented to move bulk material generally from the first end toward the second end during operation of the auger; a coned section along the auger post, the coned section comprising a first cone protruding about the periphery of the auger post, the first cone having a tapered end opposite a wide end; wherein the flighting comprises ribbon flighting and plating spanning a distance from the flighting to the first cone, in the coned section thereby forming solid auger flighting in the coned section; wherein the pitch of the flighting is greater than a diameter of the flighting wherein the coned section further comprises a second cone protruding from the auger post comprising a tapered end opposite a wide end and wherein the second cone has an orientation opposite to the first cone and wherein the wide end of the second cone abuts or is adjacent to said wide end of the first cone; and wherein a taper angle of the first cone and a taper angle of the second cone are different.
